

Smoke Test - Ecosystem Session

12.1 ECOSYSTEM - STRUCTURE AND FUNCTION

Body text at 10.8pt. Species composition and **stratification** are the two main structural features. Expressed as $\text{gm}^{-2} \text{yr}^{-1}$ for *Homo sapiens*. $\text{GPP} - \text{R} = \text{NPP}$.

12.2 Productivity

● **Gross primary productivity (GPP)** - rate of production of organic matter during photosynthesis.

12.2.1 Net primary productivity

Sub-sub-section body line.

Type	Definition	Value
GPP	Rate of organic matter production	N/A
NPP	GPP minus respiration losses	170 billion tons
Ocean NPP	Productivity of the oceans	55 billion tons
Land NPP	Remainder, on land	115 billion tons

▲ (cycle - last step feeds back to step 1)

1 **Fragmentation** - detritivores break detritus into smaller particles.

2 **Leaching** - water-soluble inorganic nutrients go down into the soil horizon.

3 **Catabolism** - bacterial and fungal enzymes degrade detritus.

4 **Humification** - accumulation of humus.

5 **Mineralisation** - humus degraded, inorganic nutrients released.

① [NOTE] All the steps in decomposition operate simultaneously on the detritus.

☆ [MEMORY AID - not in NCERT] SMOKE ONLY - Fragment, Leach, Catabolise, Humify, Mineralise.

Icon row (each must be distinct at print size):

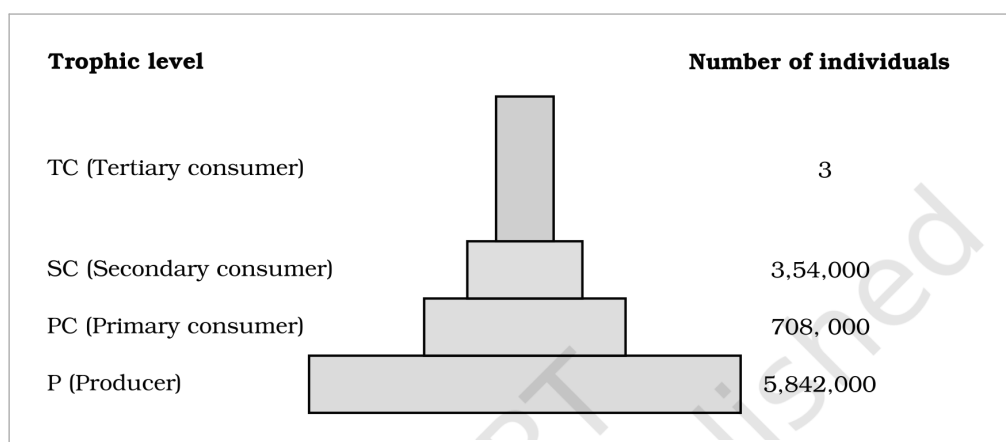


Figure 12.4 (a) Pyramid of numbers in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 millions plants